

APPENDIX A—GLOSSARY

General

STATE VARIATION: BEG-01

The following is a list of definitions of commonly used terms, in these Regulations. Definitions of terms which have their usual dictionary meanings or are used in the common technical sense, are not included.

Definitions of terms which are used solely in conjunction with radioactive materials are clearly marked "Radioactive Material Only".

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⚠ A₁ and A₂. (Radioactive Material Only).

- A₁, the activity value of Special Form radioactive material, which is listed in Table 10.3.A or derived in 10.3.2 and is used to determine the activity limits for the requirements of these Regulations.
- A₂, the activity value of radioactive material, other than Special Form radioactive material, which is listed in Table 10.3.A or derived in 10.3.2 and is used to determine the activity limits for the requirements of these Regulations.

ACIDIC. In general, an acidic substance is one which contains hydrogen and which dissolves in water to produce one or more hydrogen ions. Such solutions turn litmus dye red and cause other indicator dyes to change to characteristic colours. They also react with certain metals and bases/alkalis to form salts. Acidity is commonly measured using the pH scale; on this scale water has a "neutral" pH, i.e. neither acidic nor basic, of 7 and acids have a pH lower than 7. Some examples of *acidic* substances are hydrochloric acid, sulphuric acid, hydrogen sulphide (inorganic acids) and acetic acid, e.g. vinegar and citric acid (organic acids) (See also "Basic").

⚠ **ACTIVITY.** (Radioactive Material Only) is a measure of the quantity of radioactivity emitted by a radioisotope and is used to determine the amount of radioactive material which may be transported in various types of packagings.

AERO-ENGINES. Generic term for engines powering flying craft and fuelled by flammable liquids (e.g. petrol/gasoline, kerosene). The term includes piston designs, turbine designs and turbine auxiliary units.

AEROSOL OR AEROSOL DISPENSER. Means an article consisting of a non-refillable receptacle meeting the requirements of 6.4.4 made of metal, glass or plastic and containing a gas, compressed, liquefied or dissolved under pressure, with or without a liquid, paste or powder and fitted with a release device allowing the contents to be ejected as solid or liquefied particles in suspension in a gas, as a foam, paste or powder, or in a liquid or gaseous state.

AGENT, BLASTING, TYPE B. —see EXPLOSIVE, BLASTING TYPE B

AGENT, BLASTING, TYPE E. —see EXPLOSIVE, BLASTING TYPE E

AGGREGATE LITHIUM CONTENT. The sum of the grams of lithium content contained by the cells comprising a battery.

AIR BAG INFLATOR. —see SAFETY DEVICES, ELECTRICALLY INITIATED.

AIR BAG MODULES. —see SAFETY DEVICES, ELECTRICALLY INITIATED.

AIRCRAFT ENGINES. Generic term for engines powering flying craft fuelled by flammable liquid (jet-fuel, petrol, kerosene, etc.) which applies to piston designs, turbine designs and includes auxiliary power units (APU).

AIRLINE. —see OPERATOR.

ALKALI. —see BASIC.

ALUMINIUM POWDER. The uncoated powder may evolve hydrogen in contact with water and finely divided dust may be ignited by naked lights or sparks. Coated aluminium powders which have been treated with oils or wax for printing or paint purposes, are not generally dangerous.

ALUMINIUM PROCESSING BY-PRODUCTS. Materials, consisting of skimmings of virgin aluminium, rising to the surface of impure molten aluminium metal.

ALUMINIUM SMELTING/REMELTING BY-PRODUCTS. Materials, consisting of skimming of virgin aluminium, rising to the surface of impure molten aluminium metal.

AMMUNITION. Generic term related mainly to articles of military application consisting of all kinds of bombs, grenades, rockets, mines, projectiles and other similar devices or contrivances.

AMMUNITION, ILLUMINATING, WITH OR WITHOUT BURSTER, EXPELLING CHARGE OR PROPELLING CHARGE. Ammunition designed to produce a single source of intense light for lighting up an area. The term includes:

- Illuminating cartridges;
- Grenades;
- Projectiles;
- Illuminating and target identification bombs.

The term *excludes* Cartridges, signal; Signal devices, hand; Signals, distress; Flares, aerial and Flares, surface. These articles are listed separately.

AMMUNITION, INCENDIARY. Ammunition containing incendiary substance, which may be a solid, liquid or gel including white phosphorus. Except when the composition is an explosive *per se*, it also contains one or more of the following: a propelling charge with primer and igniter

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charge; a fuze with burster or expelling charge. The term includes:

- Ammunition, incendiary, liquid or gel, with burster, expelling charge or propelling charge;
- Ammunition, incendiary, with or without burster, expelling charge or propelling charge;
- Ammunition, incendiary, white phosphorus, with burster, expelling charge or propelling charge.

AMMUNITION, PRACTICE. Ammunition without a main bursting charge, containing a burster or expelling charge. Normally it also contains a fuze and a propelling charge. The term *excludes* Grenades, practice which is listed separately.

AMMUNITION, PROOF. Ammunition containing pyrotechnic substance(s) used to test the performance or strength of new ammunition or weapon components or assemblies.

AMMUNITION, SA (SMALL ARMS). —see CARTRIDGES, SMALL ARMS.

AMMUNITION, SMOKE. Ammunition containing smoke-producing substance such as chlorosulphonic acid mixture, titanium tetrachloride or white phosphorus. Except when the substance is an explosive *per se*, the ammunition also contains one or more of the following: a propelling charge with primer and igniter charge; a fuze with burster or expelling charge. The term includes Grenades, smoke but *excludes* Signals, smoke which are listed separately. The term includes:

- Ammunition, smoke, with or without burster, expelling charge or propelling charge;
- Ammunition, smoke, white phosphorus, with burster, expelling charge or propelling charge.

AMMUNITION, TEAR-PRODUCING, WITH BURSTER, EXPELLING CHARGE OR PROPELLING CHARGE. Ammunition containing tear-producing substance. It also contains one or more of the following:

- a pyrotechnic substance;
- a propelling charge with primer and igniter charge;
- a fuze with burster or expelling charge.

AMMUNITION, TOXIC, WITH BURSTER, EXPELLING CHARGE OR PROPELLING CHARGE. Ammunition containing toxic agent. It also contains one or more of the following:

- a pyrotechnic substance;
- a propelling charge with primer and igniter charge;
- a fuze with burster or expelling charge.

ANIMAL MATERIAL. Animal carcasses, animal body parts foodstuffs or feedstuffs derived from animals.

ANTIMONY TRICHLORIDE. A solid material which, upon exposure to air, rapidly absorbs moisture, so that its effect, in the event of leakage from a package would be the same as a solution.

APPROPRIATE NATIONAL AUTHORITY. Any authority designated, or otherwise recognized, by a State to perform specific functions related to provisions contained in these Regulations.

APPROVAL. An authorization granted by the appropriate national authority for:

- (a) the transport of dangerous goods forbidden on passenger and/or cargo aircraft where these Regulations state that goods may be carried with an approval; or
- (b) other purposes as provided for in these Regulations.

Note:

In the absence of a specific reference in these Regulations allowing the granting of an approval, an exemption may be sought.

APPROVAL. (Radioactive Material Only):

- **MULTILATERAL APPROVAL.** The approval by the relevant competent authority of the State of origin of the design or shipment, as applicable and also, where the consignment is to be transported through or into any other State, approval by the competent authority of that State. The term "through or into" specifically excludes "over", i.e. the approval and notification requirements must not apply to a State over which radioactive material is carried in an aircraft, provided that there is no scheduled stop in that State.
- **UNILATERAL APPROVAL.** The approval of a design which is required to be given by the competent authority of the State of origin of the design only.

Note:

State refers to country.

ARSENICAL DUST (ARSENICAL FLUE DUST). Consists of smelter dust containing large proportions of arsenic. These dusts are hazardous due to their toxic characteristics.

ARTICLES, EXPLOSIVE, EXTREMELY INSENSITIVE (ARTICLES, EEI). Articles that contain only extremely insensitive substances and which demonstrate a negligible probability of accidental initiation or propagation under normal conditions of transport.

Note:

An extremely insensitive substance is one which, although capable of sustaining a detonation, has demonstrated through tests that it is so insensitive that there is very little probability of accidental initiation.

ARTICLES, PYROPHORIC. Articles, which contain a pyrophoric substance, capable of spontaneous ignition when exposed to air and an explosive substance or component. The term *excludes* articles containing white phosphorus.

ARTICLES, PYROTECHNIC. For technical purposes. Articles which contain pyrotechnic substances and are used for technical purposes such as heat generation, gas generation, theatrical effects, etc. The term *excludes* Ammunition (all); Cartridges, signal; Cutters, cable, explosive; Fireworks; Flares, aerial; Flares, surface; Release devices, explosive; Rivets, explosive; Signal devices, hand; Signals, distress; Signals, railway track, explosive; Signals, smoke. These articles are listed separately.

Note:

Ammunition, cartridges, power devices, signal devices and alarm devices which contain pyrotechnic substances, are listed separately.

ASBESTOS. A generic name for naturally occurring mineral silicate fibres of the Serpentine and Amphibole series. In the Serpentine series is Chrysotile, commonly known as white asbestos. In the Amphibole series are Actinolite, Amosite, Anthophyllite, Crocidolite and Tremolite. All types of asbestos can be hazardous to health; Amphibole asbestos being the more dangerous type.

ASTM. The American Society for Testing and Materials (ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428—2959, United States of America).

AUXILIARY EXPLOSIVE COMPONENT, isolated. An "isolated auxiliary explosive component" is a small device that explosively performs an operation related to the article's functioning, other than its main explosive loads' performance. Functioning of the component does not cause any reaction of the main explosive loads contained within the article.

AVIATION REGULATED SOLID OR LIQUID. Any material which has narcotic, noxious or other properties such that in the event of spillage or leakage on an aircraft, extreme annoyance or discomfort could be caused to crew members so as to prevent the correct performance of assigned duties.

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BAGGAGE. Personal property of passengers or crew carried on an aircraft by agreement with the operator.

BAGGAGE CARRY-ON. Baggage of which the passenger or crew retains custody, also known as cabin or hand baggage.

BAGGAGE CHECKED. Baggage of which the operator takes sole custody, which is intended to be transported on the same flight(s) and for which the operator has issued a baggage check.

BAGS. Flexible packagings made of paper, plastic film, textiles, woven material or other suitable materials.

BASIC (ALKALI). In general, a basic substance dissolves in water to produce one or more hydroxyl ions. Such substances have the ability to turn litmus blue and to cause other indicators to take on characteristic colours. They also react with (neutralise) acids to form salts. Basicity is commonly measured using the pH scale; on this scale water has a "neutral" pH (neither acidic nor basic) of 7 and bases have a pH higher than 7. Some examples of *basic* substances are sodium hydroxide, (e.g. caustic soda or lye); calcium hydroxide (e.g. lime); potassium hydroxide and ammonium hydroxide. (See also "Acidic").

BATTERIES, CONTAINING SODIUM. Articles consisting of a series of Cells, containing sodium that are secured within and fully enclosed by a metal casing so constructed and closed as to prevent the release of dangerous goods under normal conditions of transport. Although designed and intended to provide a source of electrical energy, these batteries are electrically inert at any temperature at which the sodium contained in the battery is in a solid state.

BATTERIES, DRY. Sealed, non-vented batteries of the type used in flashlights or for the operation of small apparatus. They contain zinc salts and other solids, or may be of the nickel cadmium type or other combinations of metals. Such batteries must be packed in inner packagings in such a manner as to effectively prevent short circuits and to prevent movement which could lead to short circuits. Examples of such batteries are: alkali-manganese, zinc-carbon, nickel-metal hydride and nickel-cadmium batteries.

BATTERIES, DRY CONTAINING SOLID POTASSIUM HYDROXIDE. Storage batteries filled with potassium hydroxide, solid which are shipped from the factory in their original dry state and filled with the dry alkali. Water would be added to the battery before first being used.

BATTERIES, WET, ELECTRIC STORAGE. Consist of series of metal plates immersed in an electrolyte. The electrolyte is a dilute sulphuric acid, but for a certain type of battery the electrolyte is a solution of potassium hydroxide. Both of these electrolytes are corrosive liquids. The casing for the acid containing batteries is commonly plastic. Storage batteries of either of these types, when containing electrolyte, are classed as corrosive liquids. Storage batteries in transit may cause damage by leakage of the electrolyte or may produce fire by accidental short-circuiting of the terminals. Non-spillable batteries are designed and constructed so as to positively prevent leakage of the electrolyte, irrespective of the position of the battery. This is achieved by the use of jelly type electrolyte or porous absorbent separators or by specially designed filling and venting devices.

BATTERIES WET, WITHOUT ELECTROLYTE, AND FULLY DISCHARGED. Are usually wet type batteries which have been shipped from the factory in their original dry state with the intent that electrolyte would be added just before placing the batteries in service. They may also be wet batteries from which the electrolyte has been removed. In this latter instance the cells should be thoroughly flushed with water and allowed to drain before shipping.

✱ **BECCQUEREL.** (Radioactive Material Only). The becquerel is the standard unit of measure for the activity of a radionuclide used in these Regulations; it is represented by the symbol "Bq". Because the becquerel is a very small unit, larger multiples are used (see B.2.2.3). The becquerel replaces the older unit for specific activity, the "Curie" (Ci). One Ci is equal to 37 GBq.

△ **BIOLOGICAL PRODUCTS.** Are those products derived from living organisms which are manufactured and distributed in accordance with the requirements of appropriate national authorities, which may have special licensing requirements and are used either for prevention, treatment, or diagnosis of disease in humans or animals, or for development, experimental or investigational purposes related thereto. They include, but are not limited to, finished or unfinished products such as vaccines.

BLACK POWDER (GUN POWDER). Substance consisting of an intimate mixture of charcoal or other carbon and either potassium nitrate or sodium nitrate, with or without sulphur. It may be meal, granular, compressed or pelletised. Black powder may be readily ignited by a spark. The finer the grains the easier the powder may be ignited. It is

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the peculiar susceptibility to sparks which renders the transportation of black powder hazardous.

BLUE ASBESTOS. —see ASBESTOS.

BOMBS. Explosive articles which are dropped from aircraft. They may contain a flammable liquid with bursting charge, a photo-flash composition or a bursting charge. The term *excludes* Torpedoes (aerial) and includes:

- Bombs, photo-flash;
- Bombs with bursting charge;
- Bombs with flammable liquid, with bursting charge.

BOOSTERS. Articles consisting of charge of detonating explosive with or without means of ignition. They are used to increase the initiating power of detonators or detonating cord.

BOXES. Packagings with complete rectangular or polygonal faces, made of metal, wood, plywood, reconstituted wood, fibreboard, plastic or other suitable material. Small holes for purposes such as ease of handling or opening, or to meet classification requirements, are permitted as long as they do not compromise the integrity of the package during transport.

BROWN ASBESTOS. —see ASBESTOS.

△ **BUNDLES OF CYLINDERS.** (Not permitted in air transport) A pressure receptacle comprising an assembly of cylinders or cylinder shells that are fastened together and which are interconnected by a manifold and transported as a unit.

BURSTERS, EXPLOSIVE. Articles consisting of a small charge of explosive used to open projectiles, or other ammunition in order to disperse their contents.

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CALOR GAS. A liquefied flammable hydrocarbon gas or a mixture of any of the liquefiable petroleum gases.

CAPS, TOY (AMORCES). Articles consisting of a small quantity of an explosive substance between two strips or discs of paper or contained in a plastic cup or covered by varnishing or other means.

CARBON DIOXIDE. In approved cylinders, not exceeding 230 g net weight and worn by passengers for the operation of artificial limbs or orthopaedic appliances, is not considered to be a dangerous article under these Regulations, provided the cylinders are approved for the purpose and pressures do not exceed the maximum working and service pressures for which they are designed. Spare cylinders of the same size and make may be carried by the passengers if required to ensure an adequate supply for the duration of the journey.

CARBON DIOXIDE, SOLID (DRY ICE). Carbon dioxide, solid (dry ice) is produced by expanding liquid carbon dioxide to vapour and "snow" in presses that compact the product into blocks. It is used primarily for cooling and due to its very low temperature (about -79°C) can cause severe burns to skin upon direct contact. When Carbon dioxide, solid (dry ice) converts (sublimates) directly to gaseous carbon dioxide it takes in heat from its surroundings. The resulting gas is heavier than air and can cause suffocation in confined areas as it displaces air.

Packages containing Carbon dioxide, solid (dry ice) must be designed and constructed so as to prevent build-up of pressure due to the release of carbon dioxide gas.

CARGO. For the purposes of these Regulations, any property carried on an aircraft other than mail and accompanied or mishandled baggage.

CARGO AGENT. A person or organization authorized by an airline to receive shipments, execute Air Waybills and collect charges. An IATA cargo agent is one that is recognized by IATA as having met its requirements for an IATA registered cargo agent.

CARGO AIRCRAFT. Any aircraft, other than a passenger aircraft, which is carrying goods or property.

CARGO COMPARTMENT CLASSIFICATION. These definitions reflect the classification requirements set out in Federal Aviation Regulation (FAR) Section 25.857 and European Aviation Safety Agency (EASA) Certification Standard (CS) 25.857, as shown in the ICAO document Emergency Response Guidance for Aircraft Incidents Involving Dangerous Goods (Doc 9481) (red book).

Class A. A Class A cargo or baggage compartment is one in which:

- (a) the presence of a fire would be easily discovered by a crew member while at his or her station; and
- (b) each part of the compartment is easily accessible in flight.

Class B. A Class B cargo or baggage compartment is one in which:

- (a) there is sufficient access in flight to enable a crew member to effectively reach any part of the compartment with the contents of a hand fire extinguisher;
- (b) when the access provisions are being used, no hazardous quantity of smoke, flames, or extinguishing agent, will enter any compartment occupied by the crew or passengers; and
- (c) there is a separate approved smoke detector or fire detector system to give warning at the pilot or flight engineer station.

Class C. A Class C cargo or baggage compartment is one not meeting the requirements for either a Class A or B compartment but in which:

- (a) there is a separate approved smoke detector or fire detector system to give warning at the pilot or flight engineer station;
- (b) there is an approved built-in fire extinguishing or suppression system controllable from the pilot or flight engineer station;
- (c) there are means to exclude hazardous quantities of smoke, flames, or extinguishing agent, from any compartment occupied by the crew or passengers; and
- (d) there are means to control ventilation and draughts within the compartment so that the extinguishing agent used can control any fire that may start within the compartment.

Class D. A Class D cargo or baggage compartment is one in which:

- (a) a fire occurring in it will be completely confined without endangering the safety of the aeroplane or the occupants;
- (b) there are means to exclude hazardous quantities of smoke, flames, or other noxious gases from any compartment occupied by the crew or passengers;
- (c) ventilation and draughts are controlled within each compartment so that any fire likely to occur in the compartment will not progress beyond safe limits; and
- (d) consideration is given to the effect of heat within the compartment on adjacent critical parts of the aeroplane.

Class E. A Class E cargo compartment is one on aeroplanes used only for the carriage of cargo and in which:

- (a) there is a separate approved smoke or fire detector system to give warning at the pilot or flight engineer station;
- (b) there are means to shut off the ventilating airflow to, or within, the compartment and the controls for these means are accessible to the flight crew in the crew compartment;
- (c) there are means to exclude hazardous quantities of smoke, flames, or noxious gases, from the flight crew compartment; and
- (d) the required crew emergency exits are accessible under any cargo loading condition.

CARGO IMP CODES. A standard system of coding for cargo message elements. IMP codes are used by operators in data exchange in order to minimise transmission time. See B.2.2.4 for a list of Cargo IMP Codes.

CARRIER. —see OPERATOR.

CARTRIDGES. Generic term, applied to any explosive article designed to deliver combustion gases, under pressure, with a view to performing a given mechanical action, for example to propel a projectile. In particular, it applies to assembled ammunition consisting of a case fitted with a primer, filled with propellant powder with or without projectile. The term cartridge is used also to indicate a unit charge of blasting explosive, wrapped in a thin paper, plastic or other envelope, the shape of which is ordinarily a cylinder. However, cartridge-blasting explosives are not considered as articles but as substances.

CARTRIDGES, ACTUATING FOR FIRE EXTINGUISHER OR APPARATUS VALVE, EXPLOSIVE. Contrivances containing a small explosive charge with a primer, the functioning of which ruptures a metal piece (for example, a bursting disc) and thereby actuates a fire extinguisher or either opens or closes a valve (See Cartridges, power device).

CARTRIDGES, FLASH. Articles consisting of a casing, a primer and a flash powder, all assembled in one piece ready for firing.

CARTRIDGES FOR WEAPONS.

- (a) fixed (assembled) or semi-fixed (partially-assembled) ammunition designed to be fired from weapons. Each cartridge includes all the components necessary to function the weapon once. The name and description should be used for small arms cartridges that cannot be described as "Cartridges, small arms". Separate loading ammunition is included under this name and description when the propelling charge and projectile are packed together (see also "Cartridges for Weapons, blank");
- (b) incendiary, smoke, toxic and tear-producing cartridges are described in this appendix under "Ammunition, incendiary", etc.

CARTRIDGES FOR WEAPONS, BLANK. Articles which consist of a cartridge case with a centre or rim fire primer and a confined charge of smokeless or black powder but no projectile. Used for training, saluting or in starter pistols, etc.

CARTRIDGES FOR WEAPONS, INERT PROJECTILE. Ammunition consisting of a projectile without bursting charge but with a propelling charge. The presence of a tracer can be disregarded for classification purposes provided that the predominant hazard is that of the propelling charge.

CARTRIDGES, OIL WELL. Articles consisting of a casing of thin fibre, metal or other material containing only propellant which projects a hardened projectile. The term *excludes* Charges, shaped which are listed separately.

CARTRIDGES, POWER DEVICE. Articles designed to accomplish mechanical actions. They consist of a casing with a charge of deflagrating explosive and a means of ignition. The gaseous products of the deflagration produce inflation, linear or rotary motion or activate diaphragms, valves or switches or project fastening devices or extinguishing agents.

CARTRIDGES, SIGNAL. Articles designed to fire-coloured flares or other signals from signal pistols, etc.

CARTRIDGES, SMALL ARMS. Ammunition consisting of a cartridge case fitted with a centre or rim fire primer and containing both a propelling charge and solid projectile(s). They are designed to be fired in weapons of calibre not larger than 19.1 mm. Shotgun cartridges of any calibre are included in this definition. The term *excludes* Cartridges, small arms, blank which are listed separately and some small arms cartridges which are listed under Cartridges for weapons, inert projectile.

CARTRIDGES, SMALL ARMS, BLANK. —see CARTRIDGES FOR WEAPONS, BLANK.

CASES, CARTRIDGE, EMPTY, WITH PRIMER. Articles consisting of a cartridge case made from metal, plastics or other non-flammable material, in which the only explosive component is the primer.

CASES, COMBUSTIBLE, EMPTY, WITHOUT PRIMER. Articles consisting of cartridge cases made partly or entirely from nitrocellulose.

CASTOR BEANS. The residue from extraction of oil from the castor seed.

CASTOR FLAKE. —see CASTOR BEANS.

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CASTOR MEAL. —see CASTOR BEANS.

CASTOR POMACE. —see CASTOR BEANS.

CELLS, CONTAINING SODIUM. Articles consisting of hermetically sealed, metal casings which fully enclose the dangerous goods and which are so constructed and closed as to prevent the release of the dangerous goods under normal conditions of transport. In addition to sodium, cells covered by this entry may also contain sulphur, but no other dangerous goods. Although designed and intended to provide a source of electrical energy, these cells are electrically inert at any temperature at which the sodium contained in the cell is in a solid state.

CEMENT. The fine grey powder composed of lime, alumina and silica which sets to a hard product when water is added. Also known as hydraulic cement or Portland cement. It is used to make concrete. This product is not restricted for transport by air.

CEMENT, FLAMMABLE. This product, properly called an adhesive, usually contains rubber or a rubber-like substance and a solvent. It is used to bond other substances together such as paper or leather. The solvent may be flammable.

CHARGES, BURSTING. Articles consisting of a charge of detonating explosives such as hexolite, octolite or plastics bonded explosive designed to produce effect by blast or fragmentation.

CHARGES, DEMOLITION. Articles containing a charge of a detonating explosive in a casing of fibreboard, plastics, metal or other material. The term *excludes* Bombs, Mines, etc., which are listed separately.

CHARGES, DEPTH. Articles consisting of a charge of detonating explosive contained in a drum or projectile. They are designed to detonate under water.

CHARGES, EXPELLING. A charge of deflagrating explosive designed to eject the payload from the parent articles without damage.

CHARGES, EXPLOSIVE, COMMERCIAL, WITHOUT DETONATOR. Articles consisting of a charge of detonating explosive without means of initiation, used for explosive welding, jointing, forming and other metallurgical processes.

CHARGES, PROPELLING. Articles consisting of a propellant charge in any physical form, with or without a casing, for use as a component of rocket motors or for reducing the drag of projectiles.

CHARGES, PROPELLING FOR CANNON. Articles consisting of a propellant charge in any physical form, with or without a casing, for use in a cannon.

CHARGES, SHAPED WITHOUT DETONATOR. Articles consisting of a casing containing a charge of detonating explosive with a cavity lined with rigid material, without means of initiation. They are designed to produce a powerful, penetrating jet effect.

CHARGES, SHAPED, FLEXIBLE, LINEAR. Articles consisting of a V-shaped core of a detonating explosive and are clad by a flexible metal sheath.

CHARGES, SUPPLEMENTARY, EXPLOSIVE. Articles consisting of a small removable booster used in the cavity of a projectile between the fuze and the bursting charge.

CHEMICAL KITS. Boxes, cases, etc., containing small amounts of one or more compatible items of dangerous goods used for analytical or other purposes.

CHEMICAL SAMPLE, TOXIC. This entry may only be used for chemical samples taken for analysis in connection with the implementation of the *Chemical Weapons Convention*.

- ☐ **CLOSED CRYOGENIC RECEPTACLE.** A thermally insulated pressure receptacle for refrigerated liquefied gases, of a water capacity of not more than 1,000 L.

CLOSURES. Devices which close an opening in a receptacle.

- ☐ **Note:**
Examples of closures for pressure receptacles are valves, pressure relief devices, pressure gauges or level indicators.

COAL GAS COMPRESSED. The gas obtained by the destructive distillation of bituminous coal. It is shipped in steel cylinders and classed as a toxic flammable compressed gas.

COATING SOLUTIONS. Materials such as automobile undercoatings, drum or barrel lining materials, etc. which cannot properly be described as cements, but present similar hazards in transportation. They usually contain flammable solvents.

COCCULUS. Is the dried fruit or berry of an Oriental plant having toxic qualities.

COMBINATION PACKAGINGS. Are a combination of packagings for transport purposes, consisting of one or more inner packagings secured in an outer packaging in accordance with the relevant provisions of Section 5.

COMPETENCY. A dimension of human performance that is used to reliably predict successful performance on the job. A competency is manifested and observed through behaviours that mobilise the relevant knowledge, skills and attitudes to carry out activities or tasks under specified conditions.

- ✱ **COMPETENT AUTHORITY.** (Radioactive Material Only). Any body or authority designated or otherwise recognized as such for any purpose in connection with these Regulations. A list of competent authorities is given in Appendix D.2.

COMPLIANCE ASSURANCE. A systematic program of measures applied by an appropriate authority which is aimed at ensuring that the provisions of these Regulations are met in practice.

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COMPONENTS, EXPLOSIVE TRAIN, NOT OTHERWISE SPECIFIED. Devices containing an explosive, designed to transmit the detonation within an explosive train.

COMPOSITE PACKAGINGS. Packagings consisting of an outer packaging and an inner receptacle so constructed that the inner receptacle and the outer packaging form an integral packaging. Once assembled, it remains thereafter

an integrated single unit; it is filled, stored, transported and emptied as such.

Note:

Composite packagings for the purpose of these Regulations are regarded as single packagings.

☣ **CONFINEMENT SYSTEM.** (Radioactive Material Only). The assembly of fissile material and packaging components specified by the designer and agreed to by the competent authority as intended to preserve criticality safety.

CONSIGNEE. Any person, organization or government which is entitled to take delivery of a consignment.

Note:

The name and address of the consignee which appears on the Shipper's Declaration for Dangerous Goods form may differ from that on the Air Waybill.

CONSIGNMENT. One or more packages of dangerous goods accepted by an operator from one shipper at one time and at one address, receipted for in one lot and moving to one consignee at one destination address.

CONSOLIDATED SHIPMENT. —see CONSOLIDATED CONSIGNMENT.

CONSOLIDATED CONSIGNMENT. A consignment of multi-packages which has been originated by more than one person each of whom has made an agreement for carriage by air with another person other than a scheduled air carrier. Conditions applied to that agreement may or may not be the same as conditions applied by the scheduled air carrier for the same carriage.

CONSOLIDATOR. A person or organization performing a consolidation.

CONSUMER COMMODITY. A Consumer Commodity is defined as a material which is packed and distributed in a form intended or suitable for retail sales for the purposes of personal care or household use. See Special Provision A112 for Classes and Divisions permitted under this definition.

☣ **CONTAINMENT SYSTEM.** (Radioactive Material Only). The assembly of components of the packaging specified by the designer as intended to retain the radioactive material during transport.

☣ **CONTAMINATION.** (Radioactive Material Only). The presence of a radioactive substance on a surface in quantities in excess of 0.4 Bq/cm² (0.01 nCi/cm²) for beta and gamma emitters and low toxicity alpha emitters, or 0.04 Bq/cm² (0.001 nCi/cm²) for all other alpha emitters. This is either:

- *Fixed contamination*—contamination other than non-fixed contamination; or
- *Non-fixed contamination*—contamination that can be removed from a surface during normal conditions of transport.

CONTRIVANCES, WATER-ACTIVATED, WITH BURSTER, EXPELLING CHARGE OR PROPELLING CHARGE. Articles whose functioning depends upon physico-chemical reaction of their contents with water.

CONTROL TEMPERATURE. The maximum temperature at which the substance can be safely transported. It is assumed that during transport the temperature of the immediate surroundings of the package does not exceed 55°C and attains this value for a relatively short time only during each period of 24 hours.

COPRA. The dried meat of coconuts used to produce coconut oil. Copra contains up to 67% oil and is subject to spontaneous combustion.

CORD, DETONATING, FLEXIBLE. Article consisting of a core of detonating explosive enclosed in spun fabric with plastic or other covering, unless the spun fabric is sift-proof.

CORD [FUSE], DETONATING, METAL CLAD. Articles consisting of a core of detonating explosive clad by a soft metal tube with or without protective covering. When the core contains a sufficiently small quantity of explosive, the words "mild effect" are added.

CORD, IGNITER. Article consisting of textile yarns covered with black powder or another fast burning pyrotechnic composition and of a flexible protective covering; or it consists of a core of black powder surrounded by a flexible woven fabric. It burns progressively along its length with an external flame and is used to transmit ignition from a device to a charge or primer.

CRATES. Are outer packagings with incomplete sides. They are not acceptable for air transport except as an overpack.

Note:

For air transport, crates may not be used as outer packagings of composite packagings.

CREW MEMBER. A person assigned by an operator to duty on an aircraft during a flight duty period.

☣ **CRITICALITY SAFETY INDEX (CSI).** Assigned to a package, overpack or freight container containing fissile material. A number which is used to provide control over the accumulation of packages, overpacks or freight containers containing fissile material.

CRITICAL TEMPERATURE. The temperature above which the substance cannot exist in the liquid state.

CRYOGENIC LIQUIDS. Are low temperature liquefied gases, such as air, argon, helium, neon and nitrogen.

⊗

CUTTERS, CABLE, EXPLOSIVE. Articles consisting of a knife-edged device which is driven by a small charge of deflagrating explosive into an anvil.

△ **CYLINDERS.** Pressure receptacles of a water capacity not exceeding 150 L.

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D

DANGEROUS GOODS. Articles or substances which are capable of posing a hazard to health, safety, property or the environment and which are shown in the list of dangerous goods in these Regulations or which are classified according to the Regulations.

DANGEROUS GOODS ACCIDENT. An occurrence associated with and related to the transport of dangerous goods by air which results in fatal or serious injury to a person or major property or environmental damage.

DANGEROUS GOODS INCIDENT. An occurrence other than a dangerous goods accident associated with and related to the transport of dangerous goods by air, not necessarily occurring on board an aircraft, which results in injury to a person, property or environmental damage, fire, breakage, spillage, leakage of fluid or radiation or other evidence that the integrity of the packaging has not been maintained. Any occurrence relating to the transport of dangerous goods which seriously jeopardises an aircraft or its occupants is also deemed to be a dangerous goods incident. (A dangerous goods accident or incident may also constitute an aircraft accident or incident as specified in *Annex 13 to the Chicago Convention on International Civil Aviation - Aircraft Accident Investigation*).

DANGEROUS GOODS SECURITY. Measures or precautions to be taken by operators, shippers and others involved in the transport of dangerous goods aboard aircraft to minimize theft or misuse of dangerous goods that may endanger persons or property.

DE-ICING FLUIDS. Frequently contain large proportions of alcohol or other flammable liquids.

✈ **DESIGN.** (Radioactive Material Only). The description of fissile material excepted under 10.3.7.2.6, special form radioactive material, low dispersible radioactive material, package or packaging which enables such an item to be fully identified. The description may include specifications, engineering drawings, reports demonstrating compliance with regulatory requirements and other relevant documentation.

DESIGN LIFE. For composite cylinders and tubes, the maximum life (in number of years) to which the cylinder or tube is designed and approved in accordance with the applicable standard.

DESIGNATED POSTAL OPERATOR. Any governmental or non-governmental entity officially designated by a Universal Postal Union (UPU) member country to operate postal services and to fulfill the related obligations arising from the acts of the Convention on its territory.

DETONATOR ASSEMBLIES, NON-ELECTRIC, FOR BLASTING. Non-electric detonators assembled with and activated by such means as safety fuse, shock tube or detonating cord. They may be of instantaneous design or incorporate delay elements. Detonating relays incorporating detonating cord are included. Other detonating relays are included in "Detonators, non-electric".

DETONATORS. Articles consisting of a small metal or plastic tube containing explosives such as lead azide, PETN or combinations of explosives. They are designed to start a detonation train. They may be constructed to

detonate instantaneously, or may contain a delay element. The term includes:

- Detonators for ammunition and Detonators for blasting, electric, non-electric and electronic programmable;
- Detonating relays without flexible detonating cord are included.

DETONATORS, ELECTRONIC programmable for blasting. Detonators with enhanced safety and security features, utilizing electronic components to transmit a firing signal with validated commands and secure communications. Detonators of this type cannot be initiated by other means.

✈ **DOSE RATE.** Means the ambient dose equivalent or the directional dose equivalent, as appropriate, per unit time, measured at the point of interest.

DRESSING, LEATHER. May contain liquids or solvents of low flash point and hence be classified as flammable liquids.

DRUMS. Are flat-ended or convex-ended cylindrical packagings made of metal, fibreboard, plastic, plywood or other suitable materials. This definition also includes packagings of other shapes, e.g. round taper-necked packagings, or pail-shaped packagings. Jerricans are not covered by this definition.

DRY ICE. —see CARBON DIOXIDE, SOLID.

DRY SHIPPER. Insulated packagings containing refrigerated liquid nitrogen fully absorbed in a porous material and intended for transport, at low temperature, of dangerous or non-dangerous products where the design of the insulated packaging would not allow the build-up of pressure within the container and would not permit the release of any refrigerated liquid nitrogen irrespective of the orientation of the insulated packaging.

DYE, N.O.S.. Are cyclic or ring compounds, containing an amino, hydroxy, sulfonic acid, or quinone group or a combination of these groups used in the manufacture of dyes.

DYE INTERMEDIATE N.O.S.. —see DYE, N.O.S.

E

ELECTROLYTE. This term is commonly applied to the dilute sulphuric acid used in the ordinary lead plate storage batteries. The solution of potassium hydroxide used in some storage batteries is also called electrolyte. The term electrolyte is sometimes applied to the strong sulphuric acid which is meant for use in storage batteries after dilution with water.

ELEVATED TEMPERATURE SUBSTANCE. A substance which is transported or offered for transport:

- in the liquid state at a temperature at or above 100°C;
- in the liquid state with a flashpoint above 60°C and which is intentionally heated to a temperature above its flashpoint; or
- in a solid state and at a temperature at or above 240°C.

EN (standard). A European standard published by the European Committee for Standardization (CEN) (CEN-36 rue de Stassart, B-1050 Brussels, Belgium).

ENTIRE LOAD. (Explosive Material Only). Means such as substantial proportion that the practical hazard should be assessed by assuming simultaneous explosion of the whole of the explosive content of the load or package.

✱ **EXCEPTED PACKAGE.** (Radioactive Material Only). Is a packaging provision (see 10.5.8) which may be used to package radioactive material in limited quantities, instruments or articles which contain radioactive material; all of these are limited by the applicable activity limits contained in Table 10.3.C. As well, empty packages which previously contained radioactive material as limited by the requirements of 10.3.11.1.6 may be shipped under these provisions. Because the activities are so limited, categorisation, labelling and Shipper's Declarations are not required for shipments prepared under these provisions. Therefore the term "radioactive material in limited quantity" should not be confused with the "limited quantity" provisions (see Subsection 2.7) as applied to non-radioactive materials. Similarly, the "excepted package" provisions as applied to radioactive materials should not be confused with the "excepted quantity" provisions (see Subsection 2.6) applied to non-radioactive materials.

EXCEPTION. A provision in these Regulations which excludes a specific item of dangerous goods from the requirements normally applicable to that item.

EXCESS BAGGAGE (Baggage consigned as cargo). Baggage which a passenger has presented to check-in as accompanied checked baggage, but which exceeds the passenger's baggage allowance specified by the operator and which is consequently consigned as cargo in order to be sent to the same destination as the passenger.

Note:

This is commonly referred to as "unaccompanied baggage" or "baggage consigned as cargo" (see The Air Cargo Tariff (TACT) Rules 3.7.8).

✱ **EXCLUSIVE USE.** (Radioactive Material Only). The sole use, by a single shipper; of an aircraft or of a large freight container, in respect of which all initial, intermediate and final loading and unloading and shipment are carried out in accordance with the directions of the shipper or consignee, where so required by these Regulations.

EXEMPTION. Authorization, other than an approval, granted by an appropriate national authority providing relief from the provisions of these Regulations. The requirements for exemptions are given in 1.2.6.

EXPELLING CHARGE. An explosive charge designed to eject the projectile from the parent article without damage.

EXPLODE. The verb used to indicate those explosive effects capable of endangering life and property through blast, heat and projection or missiles. It encompasses both deflagration and detonation.

EXPLOSION OF THE TOTAL CONTENTS. The phrase is used in testing a single article or package or a small stack of articles or packages.

EXPLOSIVE ARTICLE. An article containing one or more explosive substances.

EXPLOSIVE, BLASTING. Detonating explosive substances used in mining, construction and similar tasks. Blasting explosives are assigned to one of five types. In addition to the ingredients listed, blasting explosives may also contain inert components such as kieselguhr, and minor ingredients such as colouring agents and stabilisers.

EXPLOSIVE, BLASTING, TYPE A. Substances consisting of liquid organic nitrates such as nitroglycerin or a mixture of such ingredients with one or more of the following: nitrocellulose, ammonium nitrate or other inorganic nitrates, aromatic nitro derivatives or combustible materials such as wood-meal and aluminium powder. These explosives must be in powdery, gelatinous or elastic form. The term includes dynamite, gelatine, blasting and gelatine dynamites.

EXPLOSIVE, BLASTING, TYPE B. Substances consisting of (a) a mixture of ammonium nitrate or other inorganic nitrates with an explosive such as trinitrotoluene, with or without other substances such as wood-meal and aluminium powder, or (b) a mixture of ammonium nitrate or other inorganic nitrates with other combustible substances which are not explosive ingredients. Such explosives must not contain nitroglycerin, similar liquid organic nitrates, or chlorates.

EXPLOSIVE, BLASTING, TYPE C. Substances consisting of either potassium or sodium chlorate or potassium, sodium or ammonium perchlorates with organic nitro derivatives or combustible materials such as wood-meal or aluminium powder or a hydrocarbon. Such explosives must not contain nitroglycerin or similar liquid organic nitrates.

EXPLOSIVE, BLASTING, TYPE D. Substances consisting of a mixture of organic nitrated compounds and combustible materials such as hydrocarbons and aluminium powder. Such explosives must not contain nitroglycerin, similar liquid organic nitrates, chlorates or ammonium nitrate. The term generally includes plastic explosives.

EXPLOSIVE, BLASTING, TYPE E. Substances consisting of water as an essential ingredient and high proportions of ammonium nitrate or other oxidizers, some or all of which are in solution. The other constituents may include nitro derivatives such as trinitrotoluene, hydrocarbons or aluminium powder. The term includes explosives, emulsion; explosive slurry and explosives, watergel.

EXPLOSIVE, DEFLAGRATING. A substance, e.g. a propellant, which reacts by deflagration rather than detonation when ignited and used in its normal manner.

EXPLOSIVE, DETONATING. A substance, which reacts by detonation rather than deflagration when, initiated and used in its normal manner.

EXPLOSIVE, EXTREMELY INSENSITIVE DETONATING SUBSTANCE (EIDS). A substance which, has demonstrated through tests that it is so insensitive that there is very little probability of accidental initiation.

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F

EXPLOSIVE, INITIATING. Explosive substances which, even in very small quantities, detonate on contact with a flame, on mild or low impact or as a result of friction; they are able to transmit detonation to other explosives close to them. The main initiating explosives are mercury fulminate and lead azide. For transport purposes some explosives, such as lead styphnate, are considered as initiating explosives because of their great sensitivity to the contact of a flame, to impact or to friction. (Both these types of sensitive explosives are referred to as primary explosives).

EXPLOSIVE, PRIMARY. An explosive substance manufactured with a view to producing a practical effect by explosion which is very sensitive to heat, impact or friction and which, even in very small quantities, either detonates or burns very rapidly. It is able to transmit detonation (in the case of initiating explosive) or deflagration to secondary explosives close to it. The main primary explosives are mercury fulminate, lead azide and lead styphnate.

EXPLOSIVE, SECONDARY. An explosive substance which is relatively insensitive (when compared to primary explosives), which is usually initiated by primary explosives with or without the aid of boosters or supplementary charges. Such an explosive may react as a deflagrating or as a detonating explosive.

EXPLOSIVE SUBSTANCE. A solid or liquid substance (or a mixture of substances) which is in itself capable by chemical reaction of producing gas at such a temperature and pressure and at such a speed as to cause damage to the surroundings. Included are pyrotechnic substances even when they do not evolve gases. A substance which is not itself an explosive but which can form an explosive atmosphere of gas, vapour or dust, is not included.

EXTERNAL CARRIAGE. Any load suspended from a helicopter or in equipment attached to a helicopter.

EXTRACTS, AROMATIC OR FLAVOURING. Consist of substances used for flavouring/odorising/aromatising food, beverages, cosmetics, etc. They may have obnoxious properties such as an overpowering odour which, in the case of a leak, may cause extreme discomfort to passengers and crew. Some may contain flammable solvents and hence have a flash point sufficiently low to require classification as flammable liquids. Others may have corrosive or toxic properties and will require appropriate classification. Note that although there is a technical difference between "extracts" and "flavourings", for the purpose of these Regulations they are treated alike under the term "extracts".

F

FILLING RATIO. The ratio of the mass of gas to the mass of water at 15°C that would fill completely a pressure receptacle ready for use.

FILMS, NITROCELLULOSE BASE. This type of film consists essentially of nitrocellulose. As such the material has a comparatively low ignition temperature and burns with great rapidity when ignited. Also, when burning, the material evolves gases that are toxic. When new and in good condition, the film is reasonably stable and free from liability to spontaneous heating and combustion. Film that

has deteriorated badly becomes quite unstable and may be liable to spontaneous heating unless kept under water.

FIRE EXTINGUISHERS. Are devices containing one or more non-flammable gases under pressure. They have a mechanism to spray the contained gas, or expel a liquid or powder through some type of nozzle.

FIRE EXTINGUISHER CHARGES. Commonly consist of packages containing sodium bicarbonate (a dry powder) which is non-hazardous and bottles containing concentrated sulphuric acid, a corrosive liquid.

FIRE LIGHTERS. Are usually made from peat, wood shavings, or sawdust and a flammable liquid.

FIRE POINT. The lowest temperature at which a liquid evolves vapour in sufficient concentration that when it is ignited in air, the liquid will continue to burn. It is usually close to the flash point.

FIREWORKS. Fireworks are pyrotechnic articles designed for entertainment.

FIRST AID KITS. Are boxes, cases, etc., containing small amounts of one or more compatible items of dangerous goods used for medical purposes.

FISSILE NUCLIDES. Uranium-233, uranium-235, plutonium-239 and plutonium-241. Fissile material is a material containing any of the fissile nuclides. Excepted from the definition of fissile material are the following:

- (a) natural uranium or depleted uranium which is unirradiated;
- (b) natural uranium or depleted uranium, which has been irradiated in thermal reactors only;
- (c) material with fissile nuclides less than a total of 0.25 g;
- (d) any combination of (a), (b) and/or (c).

These exclusions are only valid if there is no other material with fissile nuclides in the package.

FLAMMABLE. The word flammable has the same meaning as inflammable in the English language.

FLARES. Articles containing pyrotechnic substances, which are designed to illuminate, identify, signal or warn. The term includes:

- Flares, aerial;
- Flares, surface.

FLASH POINT. Is defined as the lowest temperature at which flammable vapour is given off a liquid in a test vessel in sufficient concentration to be ignited in air when exposed momentarily to a source of ignition. This does not mean the temperature at which a liquid ignites spontaneously (see Spontaneous ignition temperature).

FLASH POWDER. Pyrotechnic substance which, when ignited, produces an intense light.

FLIGHT CREW MEMBER. A licensed crew member charged with duties essential to the operation of an aircraft during a flight duty period.

FLIGHT DISPATCHER. —see Flight Operations Officer.

FLIGHT OPERATIONS OFFICER. A person designated by the operator to engage in the control and supervision of flight operations, whether licensed or not, suitably

qualified, who supports, briefs and/or assists the pilot-in-command in the safe conduct of the flight.

FRACTURING DEVICES, EXPLOSIVE, FOR OIL WELLS, WITHOUT DETONATOR. Articles consisting of a charge of detonating explosive without means of initiation. They are used to fracture the rock around a drill shaft to assist the flow of crude oil from the rock.

FREIGHT CONTAINER (Non-Radioactive Material). —see UNIT LOAD DEVICE.

✱ **FREIGHT CONTAINER.** (Radioactive Material only). An article of transport equipment designed to facilitate the transport of packaged goods by one or more modes of transport without intermediate reloading, which is of a permanent enclosed character, rigid and strong enough for repeated use and must be fitted with devices facilitating its handling, particularly in transfer between aircraft and from one mode of transport to another. In addition a small freight container is that which has an internal volume of not more than 3 m³. A large freight container is that which has an internal volume of more than 3 m³. For the transport of Class 7 material, a freight container may be used as a packaging.

FREIGHT FORWARDER. A person or organization who offers the service of arranging the transport of cargo by air.

FREQUENCY OF TESTS. The number of times a test must be carried out within a given period of time.

FUEL CELL. A fuel cell is an electrochemical device that converts the chemical energy of a fuel to electrical energy, heat and reaction products.

FUEL CELL CARTRIDGE. Fuel cell cartridge means an article that stores fuel for discharge into the fuel cell through a valve(s) that controls the discharge of fuel into the fuel cell. Fuel cell cartridges, including when contained in equipment, must be designed and constructed to prevent fuel leakage during normal conditions of transport.

FUEL CELL ENGINE. A device used to power equipment and which consists of a fuel cell and its fuel supply, whether integrated with or separate from the fuel cell and includes all appurtenances necessary to fulfil its function.

✱ **FULL LOAD.** (Radioactive Material Only). Obsolete term. See EXCLUSIVE USE.

FUSE/FUZE. Although these two words have a common origin (French fusee, fusil) and are sometimes considered to be different spellings, it is useful to maintain the convention that "fuse" refers to a cord-like igniting device whereas "fuze" refers to a device used in ammunition which incorporates mechanical, electrical, chemical or hydrostatic components to initiate a train by deflagration or detonation.

FUSE, IGNITER. Tubular, metal clad. Article consisting of a metal tube with a core of deflagrating explosive.

FUSE, SAFETY. Article consisting of a core of fine-grained black powder surrounded by a flexible woven fabric with one or more protective outer coverings. When ignited it burns at a predetermined rate without any explosive effect.

FUZES. Articles designed to start a detonation or a deflagration in ammunition. They incorporate mechanical, electrical, chemical or hydrostatic components and generally protective features. The term includes:

- Fuzes, detonating;
- Fuzes, detonating, with protective features;
- Fuzes, igniting.

G

GALLIUM. Is a silvery-white metal with a melting point of 29.7°C which may be under-cooled to almost 0°C without solidifying. It has the property of penetrating very rapidly the grain boundaries of aluminium alloys and other metals and causing embrittlement.

GAS DRIPS, HYDROCARBON. Is the liquid that condenses on compression of Pintsch Gas or the condensate from gas mains. It consists principally of a mixture of benzene and unsaturated hydrocarbon. It is very combustible and has a low flash point.

GAS TURBINE ENGINES. Generic term used for engines fuelled by flammable liquids, gas or other combustible fuels. They may power fixed wing or rotary aircraft, hover-craft (air cushion vehicles), marine vessels, land vehicles, pumps and power generating plants.

GHS. The ninth revised edition of the *Globally Harmonized System of Classification and Labelling of Chemicals*, published by the United Nations as document ST/SG/AC.10/30/Rev. 9.

GRENADES. Hand or rifle. Articles that are designed to be thrown by hand or to be projected by a rifle. The term includes:

- Grenades, hand or rifle, with bursting charge;
- Grenades, practice, hand or rifle.

The term *excludes* Grenades, smoke which are listed under "Ammunition, smoke".

GROSS WEIGHT. The total weight of the package as presented for transport.

GUN POWDER. —see BLACK POWDER.

H

HYDROCARBON GAS, COMPRESSED. Consists of hydrocarbon gas under high pressure, but not in the liquid condition.

HYDROCARBON GAS, LIQUEFIED. Consists of hydrocarbon gas from natural gas or from distillations of petroleum which are liquefied by pressure.

HYPOCHLORITE SOLUTION. Are water solutions containing a soluble hypochlorite. The concentration of the solutions vary over a wide range. The solutions are alkaline and corrosive but are not flammable. If the hypochlorite solution contacts strong acids, a decomposition takes place to produce the noxious chlorine type gases. Contact with textile fibres, etc. will cause severe damage to fibres and to colours. The solutions are used for water-treatment, bleaching, etc.

A

IAEA. The International Atomic Energy Agency (IAEA, PO Box 100—A-1400 Vienna, Austria).

□ **IAEA REGULATIONS FOR THE SAFE TRANSPORT OF RADIOACTIVE MATERIAL.** One of the editions of those Regulations, as follows:

- (a) for the 1985, 1985 (as amended 1990) editions: IAEA Safety Series No. 6;
- (b) for the 1996 edition: IAEA Safety Series No. ST-1;
- (c) for the 1996 (revised) edition: IAEA Safety Series No. TS-R-1 (ST-1, Revised);
- (d) for the 1996 (as amended 2003), 2005, 2009 editions: IAEA Safety Standards Series No. TS-R-1;
- (e) for the 2012 edition: IAEA Safety Standards Series No. SSR-6; or
- (f) for the 2018 edition: IAEA Safety Standards Series No. SSR-6 (Rev.1).

ID NUMBER. A temporary identification number (ID) in the 8000 series assigned to an article or substance for which no UN number has been assigned. (The prefix "ID" must always be used in conjunction with these numbers).

IEC. The International Electrotechnical Commission (IEC, 3, rue de Varembe, PO Box 131, CH - 1211 Geneva 20, Switzerland).

IGNITERS. Articles containing one or more explosive substances used to start deflagration in an explosive train. They may be actuated chemically, electrically or mechanically. This term *excludes* Cord, igniter; Fuse, igniter; Fuse, instantaneous, non-detonating; Fuzes, igniting; Lighters, fuse; Primers, cap type; Primers, tubular. These articles are listed separately.

IGNITION, MEANS OF. A general term used in connection with the method employed to ignite a deflagrating train of explosive or pyrotechnic substances (e.g. a primer for a propelling charge, an igniter for a rocket motor, an igniting fuze).

IMO. The International Maritime Organization (IMO, 4 Albert Embankment, London SE1 7SR, United Kingdom).

INCOMPATIBLE. Describing dangerous goods which, if mixed, would be liable to cause a dangerous evolution of heat or gas or produce a corrosive substance.

INFLAMMABLE. —see FLAMMABLE.

INITIAL BOILING POINT. The temperature at which the liquid under test first boils.

INITIATION, MEANS OF. (1) A device intended to cause the detonation of an explosive (e.g. detonator, detonator for ammunition, detonating fuze). (2) The term "with its own means of initiation" means that the contrivance has its normal initiating device assembled to it and this device is considered to present a significant risk during transport but not one great enough to be unacceptable. The term does not apply, however, to a contrivance packed together with its means of initiation provided the device is packaged so as to eliminate the risk of causing detonation of the contrivance in the event of accidental functioning of the initiating device. The means of initiating can even be assembled to the contrivance provided there are protective features such that the device is unlikely to

cause detonation of the contrivance in conditions which are associated with transport. (3) For the purposes of classification any means of initiation without two effective protective features should be regarded as Compatibility Group B; an article with its own means of initiation, without two effective protective features, would be Compatibility Group F. However, a means of initiation which itself possesses two effective protective features would be Compatibility Group D; and an article with a means of initiation which possesses two effective protective features would be Compatibility Group D or E. Means of initiation adjudged as having two effective protective features should have been approved by the appropriate national authority. A common and effective way of achieving the necessary degree of protection is to use a means of initiation which incorporates two or more independent safety features.

INNER PACKAGINGS. Are packagings for which an outer packaging is required for transport.

Note:

The "inners" of "combination packagings" are always termed "inner packagings" not "inner receptacles". A glass bottle is an example of such an "inner packaging".

INNER RECEPTACLES. Are receptacles which require an outer packaging in order to perform their containment function.

Note:

The "inners" of composite packagings are normally termed "inner receptacles". For example, the "inner" of a 6HA1 composite packaging (plastic material) is such an "inner receptacle" since it is normally not designed to perform a containment function without its "outer packaging" and is not therefore an "inner packaging".

□ **INNER VESSEL.** For a closed cryogenic receptacle, the pressure vessel intended to contain the refrigerated liquefied gas.

INORGANIC. Inorganic compounds are all elements, alloys and compounds, which do not contain a carbon-carbon linkage. Some examples of *inorganic* substances are carbides, carbon disulphide; such toxic materials as phosphene or chlorine and inorganic acids such as hydrochloric, hydrofluoric and sulphuric acids (See also ORGANIC).

INSPECTION BODY. An independent inspection and testing body approved by the appropriate national authority.

INTERMEDIATE BULK CONTAINERS (IBCs). Are rigid or flexible portable packagings that:

△ (a) Have a capacity of:

- 3,000 L or less for solids and liquids of Packing Group II and III;
- 1,500 L or less for solids of Packing Group I when packed in flexible, rigid plastics, composite, fibreboard and wooden IBCs;
- 3,000 L or less for solids of Packing Group I when packed in metal IBCs;
- 3,000 L or less for radioactive material Class 7;

- (b) Are designed for mechanical handling;
- (c) Are resistant to the stresses produced in handling and transport, as determined by tests.

Note:

IBCs are only authorised by these Regulations for UN 3077, Environmentally hazardous substances, solid, n.o.s. as provided in Packing Instruction 956.

INTERMEDIATE PACKAGINGS. Packagings placed between inner packagings, or articles and an outer packaging.

INTERNATIONAL SYSTEM OF UNITS (SI). A rational and coherent system of units which provides the basis for the units of measurement used for air and ground operations as contained in Annex 5 to the Chicago Convention on International Civil Aviation.

IRON MASS AND IRON SPONGE. Consist of a mixture of wood shavings with iron oxide and possibly lime or other material. If properly made and all the iron is properly oxidised, the material is free from the hazard of spontaneous heating or ignition. If there is an undue amount of metallic unoxidized iron, further oxidation is liable to occur, producing sufficient heat in closely packed material to cause fire. This material may be ignited by external sparks. Iron mass is used for gas purification. See IRON OXIDE, SPENT.

IRON OXIDE, SPENT OR IRON SPONGE, SPENT. Consists of iron mass or iron sponge after saturation with sulphur in gas purification. This spent material is very liable to spontaneous heating and ignition. See IRON MASS AND IRON SPONGE.

ISO (standard). An international standard published by the International Organization for Standardization (ISO—1, Voie-Creuse, CH-1211 Geneva 20, Switzerland).

ISOCYANATES. Include a number of chemical products used in the manufacture of plastic foams, synthetic rubber, etc. Some are sufficiently toxic or lachrymatory to need classification as toxic substances, particularly isocyanates in pure form. Others may need to be classified as flammable liquids, depending on their characteristics and a number may not be subject to these Regulations.

ISOSORBIDE DINITRATE MIXTURE. Requires the use of a phlegmatizer such as lactose, mannose, calcium, hydrogen phosphate or starch. However, other phlegmatizers may be used at the discretion of the appropriate authority, provided they have identical phlegmatizing properties.

J

JERRICANS. Are metal or plastic packagings of rectangular or polygonal cross-section.

JET PERFORATING GUNS, OIL WELL, WITHOUT DETONATOR. Article consisting of a steel tube or metallic strip into which are inserted shaped charges connected by detonating cord, without means of ignition.

L

LACQUER BASE OR LACQUER CHIPS, DRY. May consist of a colloided solid mixture of nitrocellulose, pigment, gums and a plasticizer. Those containing nitrocellulose are highly flammable.

LARGE PACKAGINGS. Are packagings consisting of an outer packaging which contains articles or inner packagings and which are designed for mechanical handling and exceed 400 kg net mass or 450 L capacity but have a volume not more than 3,000 L.

Note:

Large packagings are only permitted as provided for in 5.0.1.12 and S-4; 13 of the Supplement to the Technical Instructions.

LARGE SALVAGE PACKAGING. (Not permitted in air transport) A special packaging which:

- (a) is designed for mechanical handling; and
- (b) exceeds 400 kg net mass or 450 L capacity but has a volume of not more than 3 m³;

into which damaged, defective, leaking or non-conforming dangerous goods packages, or dangerous goods that have spilled or leaked are placed for purposes of transport for recovery or disposal. They are not permitted in air transport.

LIGHTERS, FUSE. Articles of various design actuated by friction, percussion or electricity and used to ignite safety fuse.

LINER. A separate tube or bag inserted into a packaging but not forming an integral part of it, including the closures of its openings.

△ **LIQUID.** A substance classified as dangerous goods which at 50°C has a vapour pressure of not more than 300 kPa (3 bar), which is not completely gaseous at 20°C and at a pressure of 101.3 kPa and which has a melting point or initial melting point of 20°C or less at a pressure of 101.3 kPa. A viscous substance for which a specific melting point cannot be determined must be subjected to the ASTM D 4359-90 test; or to the test for determining fluidity (penetrometer test) prescribed in section 2.3.4 of Annex A of the Agreement concerning the International Carriage of Dangerous Goods by Road (ADR) (UN Publication ECE/TRANS/300 (Sales No. E21.XIII.1)).

LITHIUM BATTERIES INSTALLED IN CARGO TRANSPORT UNIT. Typically, this applies to lithium ion batteries installed into multi-modal shipping containers (cargo transport unit) where the completed unit acts as a large storage battery. The completed unit will contain lithium ion batteries plus battery management systems and may include air conditioning and a fire suppression system.

LITHIUM BATTERY. "Battery" means two or more cells which are electrically connected together and fitted with devices necessary for use, for example, case, terminals, marking and protective devices. A single cell lithium battery is considered a "cell" and must be tested according to the testing requirements for "cells" for the purposes of these Regulations and the provisions of subsection 38.3 of the UN Manual of Tests and Criteria (see also the definition for "lithium cell").

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Note:

Units that are commonly referred to as "battery packs", "modules" or "battery assemblies" having the primary function of providing a source of power to another piece of equipment are for the purposes of these Regulations and the provisions of Subsection 38.3 of the UN Manual of Tests and Criteria treated as batteries.

The term "lithium battery" refers to a family of different chemistries, comprising many types of cathodes and electrolytes. For the purposes of the Regulations they are separated into:

- **Lithium metal batteries.** Are normally primary (non-rechargeable) batteries that have lithium metal or lithium compounds as an anode. The most common type of lithium cell used in consumer applications uses metallic lithium as anode and manganese dioxide as cathode, with a salt of lithium dissolved in an organic solvent; and
- **Lithium-ion batteries** (sometimes abbreviated Li-ion batteries). Are a type of secondary (rechargeable) battery commonly used in consumer electronics. Also included within lithium-ion batteries are lithium polymer batteries.

LITHIUM CELLS. A single encased electrochemical unit (one positive and one negative electrode) which exhibits a voltage differential across its two terminals, and may contain protective devices. Under these Regulations and the UN Manual of Tests and Criteria, to the extent the encased electrochemical unit meets the definition of "cell" herein, it is a "cell", not a "battery", regardless of whether the unit is termed a "battery" or a "single cell battery" outside of these Regulations and the UN Manual of Tests and Criteria.

LITHIUM SILICON. Is a so-called alloy of metallic lithium and silicon used for industrial purposes. The material is somewhat combustible and reactive with water. The containers must be maintained in a condition to prevent entrance of moisture.

- ⚡ **LOW DISPERSIBLE RADIOACTIVE MATERIAL.** (Radioactive Material Only). A solid radioactive material or a solid radioactive material in a sealed capsule, that has limited dispersibility and is not in powder form.
- ⚡ **LOW SPECIFIC ACTIVITY (LSA) MATERIAL.** Radioactive material which by its nature has a limited specific activity, or radioactive material for which limits of estimated average activity apply.
- ⚡ **LOW TOXICITY ALPHA EMITTER.** (Radioactive Material Only). Natural uranium; depleted uranium; natural thorium; uranium-235 or uranium-238; thorium-232; thorium-228 and thorium-230 when contained in ores or physical or chemical concentrates; and radionuclides with a half-life of less than 10 days.

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MAGNESIUM SCRAP. Is borings, clippings, shavings, sheets, turnings, or scalplings from machining operations or cuttings from thin magnesium metal sheets. The scrap can be ignited by external flame and burns intensely and persistently. It does not heat spontaneously. The scrap may have a bright metal lustre or may be dull and sometimes have a painted surface.

MAGNETIZED MATERIALS. Cover materials with relatively high magnetic field strength such as magnetrons and non-shielded permanent magnets without keeper bars installed. Masses of ferro-magnetic metals such as automobiles, automobile parts, metal fencing, piping and metal construction material, even if not meeting the definition of magnetized materials, may be subject to operator's special stowage regulations since they may affect aircraft instruments, particularly the compasses.

MAIL. Dispatches of correspondence and other items tendered by and intended for delivery to postal services in accordance with the rules of the Universal Postal Union (UPU).

⚡ **MANAGEMENT SYSTEM.** A set of interrelated or interacting elements (system) for establishing policies and objectives and enabling the objectives to be achieved in an efficient and effective manner.

□ **MANUAL OF TESTS AND CRITERIA.** The seventh revised edition of the United Nations publication bearing this title (ST/SG/AC.10/11/Rev.7 and Amend. 1).

MASS. —see WEIGHT.

MASS EXPLOSION. An explosion that affects almost the entire load virtually instantaneously.

MATCHES, SAFETY (BOOK, CARD, OR STRIKE-ON-BOX). Are matches intended to be struck on a prepared surface.

MATCHES, STRIKE-ANYWHERE OR FUSEE. Usually contain phosphorus sesquisulphide, potassium chlorate and other ingredients. The strike-anywhere matches are readily ignited by friction on almost any dry surface. When a closed package of strike-anywhere matches is ignited by impact or friction the head composition burns off the matches and the fire then usually goes out unless the package is broken. If the package is broken, allowing access of air, the fire will continue. Packages of these matches that have been wetted for any reason and subsequently dried should be handled with extreme caution.

MAXIMUM CAPACITY. As used in Section 6 is the maximum inner volume of receptacles or packagings expressed in litres.

MAXIMUM NET MASS. As used in Section 6 is the maximum net mass of contents in a single packaging or maximum combined mass of inner packagings and the contents thereof expressed in kilograms.

⚡ **MAXIMUM NORMAL OPERATING PRESSURE.** (Radioactive Material Only). The maximum pressure above atmospheric pressure at mean sea-level that would develop in the containment system in a period of one year under the conditions of temperature and solar radiation corresponding to environmental conditions of transport in the absence of venting, external cooling by an ancillary system or operational controls during transport.

MAXIMUM TRANSPORT TEMPERATURE. The maximum temperature likely to be encountered by a substance in transport.

METAL CATALYST, FINELY DIVIDED, ACTIVATED, OR SPENT. Is metal in an extremely finely divided form. It must be shipped wet in moisture-tight and airtight packagings. If exposed to air, the metal may become hot and may even ignite.

△ **METAL HYDRIDE STORAGE SYSTEM.** A single complete hydrogen storage system, including a pressure receptacle shell, metal hydride, pressure relief device, shut-off valve, service equipment and internal components used for the transport of hydrogen only.

METHYLACETYLENE AND PROPADIENE MIXTURE, STABILIZED. A flammable gas mixture that is reasonably stable at ordinary temperatures. While this is an acetylene derivative, the gas is not shipped dissolved in liquid and the cylinders do not require an absorbent filler.

⊗ **MINES.** Articles normally consisting of metal or composition receptacles and a bursting charge. They are designed to be operated by the passage of ships, vehicles or personnel. The term includes "Bangalore torpedoes".

MISCIBLE. The ability of a liquid (or gas) to dissolve uniformly in another liquid (or gas). Miscibility depends on the chemical nature of the substances involved and in some cases, liquids may only be partially miscible. Liquids that do not mix at all are said to be *immiscible*.

MIXTURE. Mixture means a material composed of more than one chemical compound or element.

□ **MODEL REGULATIONS.** The twenty-second revised edition of the United Nations publication entitled Recommendations on the Transport of Dangerous Goods, Model Regulations (ST/SG/AC.10/1/Rev.22).

MOTOR FUEL ANTI-KNOCK MIXTURE. A mixture of one or more organic lead compounds such as tetraethyl lead, triethylmethyl lead, diethyldimethyl lead, ethyltrimethyl lead and tetramethyl lead, with one or more halogen compounds such as ethylene dibromide and ethylene dichloride.

✈ **MULTILATERAL APPROVAL.** (Radioactive Material Only). The approval of a design by the competent authorities of the State of origin and of each State through or into which (see Note) the consignment is to be transported.

Note:

The term "through or into" specifically excludes "over", i.e. the approval and notification requirements would not apply to a State over which radioactive material is carried in an aircraft, provided that there is no scheduled stop in that State.

MULTIPLE-ELEMENT GAS CONTAINERS (MEGCs). (Not permitted for air transport.) Are multi modal assemblies of cylinders, tubes and bundles of cylinders which are interconnected by a manifold and which are assembled within a framework. The MEGC includes service equipment and structural equipment necessary for the transport of gases.

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NET EXPLOSIVE MASS (NEM). The total mass of the explosive substances, without the packagings, casings, etc. (net explosive quantity (NEQ), net explosive contents (NEC), or net explosive weight (NEW) are often used to convey the same meaning).

NET QUANTITY. Either:

- the weight or volume of the dangerous goods contained in a package excluding the weight or volume of any packaging material; or
- the weight of an unpackaged article of dangerous goods (e.g. UN 3166)

For the purposes of this definition "dangerous goods" means the substance or article as described by the proper shipping name shown in Table 4.2, e.g. for "Fire extinguishers", the net quantity is the weight of the fire extinguisher. For articles packed with equipment or contained in equipment, the net quantity is the net weight of the article, e.g. for "Lithium ion batteries contained in equipment", the net quantity is the net weight of the lithium ion batteries in the package.

NEUTRON RADIATION DETECTOR. A device that detects neutron radiation. In such a device, a gas may be contained in a hermetically sealed electron tube transducer that converts neutron radiation into a measurable electric signal.

NITRATING ACID MIXTURE. A mixture of nitric and sulphuric acids used for the nitration of glycerine, cellulose or other organic substances. This acid mixture coming in contact with organic matter commonly causes fire, unless the mixture contains much water.

NOT RESTRICTED. Means not subject to or restricted by these Regulations, except as otherwise stated.

O

OIL GAS COMPRESSED. A gas made by the reaction of steam at high temperatures on gas oil or similar fractions of petroleum, or by high temperature cracking of gas oil. The gas is flammable but is classified as a toxic gas because it contains a high proportion of Carbon Monoxide.

OPEN CRYOGENIC RECEPTACLE. A transportable thermally insulated receptacle for refrigerated liquefied gases maintained at atmospheric pressure by continuous venting of the refrigerated liquefied gas.

OPERATOR. A person, organisation or enterprise engaged in or offering to engage in an aircraft operation.

ORGANIC. Organic substances are compounds which contain a carbon-carbon linkage. There are a few exceptions but by methodical reasons they are counted to belong to the organic compounds (e.g. Methane CH₄), for example carbon oxides, the carbides, carbon disulphide, etc.; ternary compounds such as metallic cyanides, metallic carbonyls, phosgene (COCl₂), carbonyl sulphide (COS), etc.; and the metallic carbonates, such as calcium carbonate and sodium carbonate. Some examples of *organic* substances are hydrocarbons, alcohols, ethers, aldehydes, ketones, organic solvents and organic acids (See also "Inorganic").

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ORGANIC PEROXIDES. Organic peroxides are classified into seven types according to the degree of danger. These types range from A, which is not accepted for transport, to G which is exempted from Division 5.2. The classification of B to F is directly related to the maximum quantity allowed in one package. The list of currently assigned Organic Peroxides is shown in Appendix C.

OUTER PACKAGING. The outer protection of a composite or combination packaging together with any absorbent materials, cushioning and any other components necessary to contain and protect inner receptacles or inner packagings.

OVERPACK. An enclosure used by a single shipper to contain one or more packages and to form one handling unit for convenience of handling and stowage. Dangerous goods packages contained in the overpack must be properly packed, marked, labelled and in proper condition as required by these Regulations. For cooling purposes, an overpack may contain Carbon dioxide, solid (dry ice), provided that the overpack meets the requirements of Packing Instruction 954. (A Unit Load Device is not included in this definition).

Note:

Shrink-wrap or banding may be considered an overpack.

OXYGEN GENERATOR, CHEMICAL. Devices containing chemicals which upon activation release oxygen as a product of chemical reaction. Chemical oxygen generators are used for the generation of oxygen for respiratory support, e.g. in aircraft, submarines, spacecraft, bomb shelters and breathing apparatus. Oxidizing salts such as chlorates and perchlorates of lithium, sodium and potassium, which are used in chemical oxygen generators, evolve oxygen when heated. These salts are mixed (compounded) with a fuel, usually iron powder, to form a chlorate candle, which produces oxygen by continuous reaction. The fuel is used to generate heat by oxidation. Once the reaction begins, oxygen is released from the hot salt by thermal decomposition (a thermal shield is used around the generator). A portion of oxygen reacts with fuel to produce more heat, which produces more oxygen and so on. Initiation of the reaction can be achieved by a percussion device, friction device or electric wire.

P

PACKAGE. (Non-Radioactive Material). The complete product of the packing operation consisting of the packaging and contents prepared for transport.

PACKAGING. (Non-Radioactive Material). One or more receptacles and any other components or materials necessary for the receptacles to perform their containment and other safety functions and to ensure compliance with the minimum packing requirements of these Regulations.

⚠ **PACKAGING.** (Radioactive Material Only). The assembly of components necessary to enclose completely the radioactive contents. It may in particular, consist of one or more receptacles, absorbent materials, spacing structures, radiation shielding, service equipment for filling, emptying, venting and pressure relief and devices integral to the package. The packaging may be a box, drum, or similar receptacle, or may also be a freight container.

PACKING. The art and operation by which articles or substances are enveloped in wrappings and/or enclosed in packagings or otherwise secured.

PACKING GROUP. An indication of the relative degree of danger presented by substances within a class or division. Roman numerals I, II and III are used to represent "high danger", "medium danger" and "low danger" respectively.

PASSENGER AIRCRAFT. An aircraft that carries any person other than a crew member, an operator's employee in an official capacity, an authorised representative of an appropriate national authority or a person accompanying a consignment or other cargo.

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PERCENTAGE FILLED. The percentage of the capacity filled with the prescribed material.

PHLEGMATISER. A solid or liquid which is added to a substance such as an explosive or an organic peroxide in order to reduce its sensitivity to heat and impact, thereby assisting in ensuring safety during transport.

PHLEGMATIZED. Means that a substance (or "phlegmatizer") has been added to an explosive to enhance its safety in handling and transport. The phlegmatizer renders the explosive insensitive, or less sensitive, to the following actions: heat, shock, impact, percussion or friction. Typical phlegmatizing agents include, but are not limited to: wax, paper, water, polymers (such as chloro-fluoropolymers), alcohol and oils (such as petroleum jelly and paraffin).

PHOTO-FLASH POWDER, IN UNITS. —see FLASH POWDER.

PILOT-IN-COMMAND. The pilot designated by the operator, or in the case of general aviation, the owner, as being in command and charged with the safe conduct of the flight.

PLASTIC SOLVENT. A name commonly used for mixtures of liquids employed for dissolving plastics or for thinning plastic cements. In general, they may contain flammable liquids, such as acetone, amyl acetate, or some of the alcohols or ketones. The classification is determined by the flash point.

POLYESTER RESIN KIT. The proper shipping name "Polyester resin kit" covers different kits such as filler, bonding and sealing compounds, chemical anchors and fibreglass repair kits. A polyester resin kit commonly consists of an unsaturated polyester resin mixed with styrene and a separate hardener (usually a phlegmatized organic peroxide) as a minor component. The main component (viscous liquid or paste) is inherently flammable due to the styrene content (flash point 29°C–32°C).

POLYMERIC BEADS, EXPANDABLE. Semi-processed products used to manufacture polymeric articles. These are impregnated with a flammable gas or liquid as a blowing agent and may evolve small quantities of flammable gas during transportation.

POLYMERIZABLE MATERIAL. Any liquid, solid, or gaseous material which, under conditions incident to transportation, may polymerize (combine or react with itself) so as to cause dangerous evolution of gas or heat.

PORTABLE TANK. A tank having a capacity of more than 450 L whose shell is fitted with items of service equipment and structural equipment necessary for the transport of dangerous goods, that has stabilising members external to the shell and that is not permanently secured aboard the aircraft. It must be capable of being filled and discharged without the need for the removal of its structural equipment and must be capable of being lifted on and off the aircraft when filled.

POST. See MAIL.

POTASSIUM SODIUM ALLOYS. Are mixtures of metallic sodium and potassium that are solid at ordinary temperatures. All mixtures, regardless of physical state, will react vigorously with water and may be self-igniting. The mixtures are all combustible.

POTASSIUM SULPHIDE, ANHYDROUS. A reddish-coloured solid having a strong odour. It is hygroscopic and oxidizes spontaneously on contact with air. Spontaneous ignition may occur in material improperly packed.

POWDER CAKE, WETTED. Substances consisting of nitrocellulose impregnated with not more than 60% of nitroglycerin or other liquid organic nitrates or a mixture of these.

POWDER PASTE, WETTED. —see POWDER CAKE, WETTED.

POWDER, SMOKELESS. Substances based on nitrocellulose used as propellant. The term includes propellants with a single base (nitrocellulose [NC] alone), those with a double base (such as NC and nitroglycerin [NG]) and those with a triple base (such as NC/NG/nitroguanidine). Cast, pressed or bag-charges of smokeless powder are listed under "Charges, propelling" or "Charges, propelling for cannon".

PREMIXING BURNER LIGHTER. Gas lighter in which fuel and air are mixed before being supplied for combustion, such as lighters producing a blue flame.

Note:

Premixing burner ("blue flame") lighters are forbidden for carriage by passengers and crew.

△ PRESSURE DRUMS. (Not permitted for air transport). Are welded pressure receptacles of a water capacity exceeding 150 L and of not more than 1,000 L, (e.g. cylindrical receptacles equipped with rolling hoops, spheres on skids).

△ PRESSURE RECEPTACLE. A transportable receptacle intended for holding substances under pressure including its closure(s) and other service equipment and a collective term that includes cylinders, tubes, pressure drums, closed cryogenic receptacles, metal hydride storage systems and bundles of cylinders.

□ PRESSURE RECEPTACLE SHELL. A cylinder, a tube, a pressure drum or a salvage pressure receptacle without its closures or other service equipment, but including any permanently attached device(s) such as a neck ring or a foot ring.

Note:

The terms "cylinder shell", "pressure drum shell" and "tube shell" are also used.

PRIMERS, CAP TYPE. Articles consisting of metal or plastics caps containing a small amount of primary explosive mixture that is readily ignited by impact. They serve as igniting elements in small arms cartridges and in percussion primers for propelling charges.

PRIMERS, TUBULAR. Articles consisting of a primer for ignition and an auxiliary charge of deflagrating explosive such as black powder used to ignite the propelling charge in a cartridge case for cannon, etc.

PROJECTILES. Articles such as a shell or bullet which are projected from a cannon or other artillery gun, rifle or other small arm. They may be inert, with or without tracer, or may contain burster or expelling charge or a bursting charge. The term includes:

- Projectiles, inert, with tracer;
- Projectiles, with burster or expelling charge;
- Projectiles, with bursting charge.

PROPELLANTS. Deflagrating explosives used for propulsion or for reducing the drag of projectiles.

PROPELLANT, LIQUID. A substance consisting of a deflagrating liquid explosive, used for propulsion or for reducing the drag of projectiles.

PROPELLANT, SOLID. A substance consisting of a deflagrating solid explosive, used for propulsion or for reducing the drag of projectiles.

PROPER SHIPPING NAME. The name to be used to describe a particular article or substance in all shipping documents and notifications and where appropriate, on packagings.

Note:

These names are indicated in bold face characters in the List of Dangerous Goods. See also 8.1.3.

PROTECTIVE BREATHING EQUIPMENT (PBE). Also known as a "smoke hood", PBE are carried on commercial aircraft as part of the safety equipment for flight and cabin crew. PBE contain a small chemical oxygen generator (UN 3356) similar to that found fitted in the cabin for passenger use.

PYROPHORIC LIQUID/SOLID, ORGANIC/INORGANIC. A substance that may ignite in air at or below room temperature in the absence of added heat, shock or friction. All are decidedly combustible and all fume strongly on exposure to air to produce fumes that are somewhat irritating and may be somewhat toxic.

PYROTECHNIC SUBSTANCE. A mixture or compound designed to produce an effect by heat, light, sound, gas or smoke, or a combination of these, as the result of non-detonative, self-sustaining, exothermic, chemical reactions.

PYROXYLIN SOLUTION. Consists of pyroxylin (nitrocellulose) or soluble cotton dissolved in amyl acetate or other organic solvents. Pyroxylin solution is used as a basis for the manufacture of lacquer, leather coating compounds, leather substitutes, cements, etc. It is generally more viscous than ordinary lacquers.

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QUALITY ASSURANCE. A systematic program of controls and inspections applied by any organization or body, which is aimed at providing adequate confidence that the standard of safety prescribed by these Regulations is achieved in practice.

QUALITY CONTROL. Several production samples selected at random must be tested in accordance with the requirements and frequency described for each type of package. If the requirements are not met, the whole production lot must be rejected unless otherwise specified.

QUANTITY. For packaging tests, quantity means a certain number of random samples, or in certain cases all of each individual type and size, out of a lot or out of a production series.

R

RADIATION DETECTION SYSTEM. An apparatus that contains radiation detectors as components.

RADIOACTIVE CONTENTS. (Radioactive Material Only). The radioactive material together with any contaminated solids, liquids and gases within the packaging.

RECEPTACLE. A containment vessel, including closures, for receiving and holding substances or articles.

RECEPTACLE. (Explosive Material Only). Inner and intermediate packagings including boxes, bottles, cans, drums, jars and tubes, including any means of closure.

RE-CONDITIONED PACKAGINGS.

(a) Include metal drums that are:

1. cleaned to original materials of construction, with all former contents, internal and external corrosion and all external coatings and labels removed;
2. restored to original shape and contour, with chimes (if any) straightened and sealed and all non-integral gaskets replaced;
3. inspected after cleaning but before painting, with rejection of packagings with visible pitting, significant reduction in material thickness, metal fatigue, damaged threads or closures, or with other significant defects.

(b) Include plastic drums and jerricans that:

1. are cleaned to original materials of construction, with all former contents, external coatings and labels removed;
2. have all non-integral gaskets replaced; and
3. are inspected after cleaning with rejection of packagings with visible damage such as tears, creases or cracks, or damaged threads or closures or other significant defects.

RECYCLED PLASTIC MATERIAL. Material recovered from used industrial packagings that has been cleaned and prepared for processing into new packagings. The specific properties of the recycled material used for production of new packagings must be assured and documented regularly as part of a quality assurance

program recognized by the appropriate national authority. The quality assurance program must include a record of proper re-sorting and verification that each batch of recycled plastic material has the proper melt flow rate, density and tensile yield strength, consistent with that of the design type manufactured from such recycled material. This necessarily includes knowledge about the packaging material from which the recycled plastic have been derived, as well as awareness of the prior contents of those packagings if those prior contents might reduce the capability of new packagings produced using that material. In addition, the packaging manufacturer's quality assurance program must include performance of the mechanical design type tests in Subsection 6.3 on packagings manufactured from each batch of recycled plastic material. In this testing, stacking performance may be verified by appropriate dynamic compression testing rather than static load testing.

△ Note:

ISO 16103:2005 "Packaging—Transport packages for dangerous goods—Recycled plastic material", provides additional guidance on procedures to be followed in approving the use of recycled plastic material. These guidelines have been developed based on the experience of the manufacturing of drums and jerricans from recycled plastics material and as such may need to be adapted for other types of packagings, IBCs and large packagings made of recycled plastics material.

REELS. (Explosive Material Only). Devices made of plastics, wood, fibreboard, metal or other suitable material comprising a central spindle with, or without, side walls at each end of the spindle. Articles and substances can be wound onto the spindle and may be retained by side walls.

RELEASE DEVICES, EXPLOSIVE. Articles consisting of a small charge of explosive with means of initiation. They sever rods or links to release equipment quickly.

REMANUFACTURED LARGE PACKAGING. (Not permitted for air transport) Means a metal or rigid plastic large packaging that:

- (a) is produced as a UN type from a non-UN type; or
- (b) is converted from one UN design type to another UN design type.

Remanufactured large packagings are subject to the same requirements of these Regulations as apply to new large packagings of the same type.

REMANUFACTURED PACKAGINGS.

(a) Include metal drums that:

1. are produced as a UN type from a non-UN type;
2. are converted from one UN type to another UN type; or
3. undergo the replacement of integral structural components (such as non-removable heads).

(b) Include plastic drums that:

1. are converted from one UN type to another UN type (e.g. 1H1 to 1H2); or
2. undergo the replacement of integral structural components.

Remanufactured drums are subject to the same requirements of these Regulations as apply to a new drum of the same type.

REUSED LARGE PACKAGING. (Not permitted for air transport.) A large packaging to be refilled which has been examined and found free of defects affecting the ability to withstand the performance tests: the term includes those which are refilled with the same or similar compatible contents and are transported within distribution chains controlled by the consignor of the product.

REUSED PACKAGINGS. Are packagings to be filled which have been examined and found free of defects affecting the ability to withstand the performance tests; the term includes those which are transported within distribution chains controlled by the shipper of the product.

ROCKET MOTORS. Articles consisting of a solid, liquid or hypergolic fuel contained in a cylinder fitted with one or more nozzles. They are designed to propel a rocket or a guided missile. The term includes:

- Rocket motors;
- Rocket motors with hypergolic liquids, with or without expelling charge;
- Rocket motors, liquid fuelled.

ROCKETS. Articles consisting of a rocket motor and a payload which may be an explosive warhead or other device. The term includes "Guided Missiles" and:

- Rockets, line-throwing;
- Rockets, liquid-fuelled, with bursting charge;
- Rockets, with bursting charge;
- Rockets, with expelling charge;
- Rockets, with inert head.

S

SADT. —see SELF-ACCELERATING DECOMPOSITION TEMPERATURE.

SAFETY DEVICES, ELECTRICALLY INITIATED. Articles which contain pyrotechnical substances or dangerous goods of other classes and are used in vehicles, vessels or aircraft to enhance safety to persons. Examples are: air bag inflators, air bag modules, seat-belt pretensioners and pyromechanical devices. These pyromechanical devices are assembled components for tasks such as but not limited to separation, locking, or release-and-drive or occupant restraint. The term includes **Safety devices, pyrotechnic.**

SALVAGE PACKAGINGS. Special packagings into which damaged, defective, leaking or non conforming dangerous goods packages, or dangerous goods that have spilled or leaked, are placed for purposes of transport for recovery or disposal.

SALVAGE PRESSURE RECEPTACLE. (Not permitted for air transport) Means a pressure receptacle with a water capacity not exceeding 1,000 L into which are placed damaged, defective, leaking or non-conforming pressure receptacle(s) for the purpose of transport e.g. for recovery or disposal.

SEAT-BELT PRETENSIONER. Articles which contain pyrotechnical substances and are used as life-saving vehicle seatbelts.

SELF-ACCELERATING DECOMPOSITION TEMPERATURE (SADT). The lowest temperature at which self-accelerating decomposition may occur in a substance in the packaging, IBC or portable tank as offered for transport. The SADT must be determined in accordance with the test procedures given in Part II, Section 28 of the *UN Manual of Tests and Criteria*.

Note:

IBC and portable tanks are not permitted in air transport unless otherwise provided for in these Regulations.

SELF-ACCELERATING POLYMERIZATION TEMPERATURE (SAPT). The lowest temperature at which self-accelerating polymerization may occur with a substance in the packaging, IBC or portable tank as offered for transport. The SAPT must be determined in accordance with the test procedures established for the self-accelerating decomposition temperature for self-reactive substances in accordance with Part II, Section 28 of the *UN Manual of Tests and Criteria*.

Note:

IBC and portable tanks are not permitted in air transport unless otherwise provided for in these Regulations.

SERIOUS INJURY. An injury which is sustained by a person in an accident and which:

- (a) requires hospitalisation for more than 48 hours, commencing within seven days from the date the injury was received; or
- (b) results in a fracture of any bone (except simple fractures of fingers, toes or nose); or
- (c) involves lacerations which cause severe haemorrhage, nerve, muscle or tendon damage; or
- (d) involves injury to any internal organ; or
- (e) involves second or third degree burns, or any burns affecting more than 5% of the body surface; or
- (f) involves verified exposure to infectious substances or injurious radiation.

☐ **SERVICE EQUIPMENT.** For pressure receptacles, includes:

- (a) closure(s);
- (b) manifold(s);
- (c) piping;
- (d) porous, absorbent or adsorbent material; and
- (e) any structural devices such as those used for handling.

SERVICE LIFE. For composite cylinders and tubes, the number of years the cylinder or tube is permitted to be in service.

SETTLED PRESSURE. The pressure of the contents of a pressure receptacle in thermal and diffusive equilibrium.

SHIPMENT. The specific movement of a consignment from origin to destination.

⚠ **SIEVERT.** (Radioactive Material Only). The sievert is the standard unit of measure for radiation dose-equivalent used in these Regulations; it is represented by the symbol

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S

"Sv". Other multiples of the sievert are frequently used in these Regulations (see B.2.2.3). The sievert replaces the older unit for dose-equivalent, the "rem". One Sv is equal to 100 rem.

SIFT-PROOF PACKAGINGS. Packagings impermeable to dry contents including fine solid material produced during transport.

SIGNALS. Articles containing pyrotechnic substances designed to produce signals by means of sound, flame or smoke or any combinations thereof. The term includes:

- Signal devices, hand;
- Signals, distress, ship;
- Signals, railway track, explosive;
- Signals, smoke.

SINGLE PACKAGINGS. Are packagings which do not require any inner packaging in order to perform their containment function during transport.

SLUDGE, ACID. The acid waste resulting from oil refining, or from nitrating processes. It generally has somewhat the same hazards as the original acid.

SODA LIME. A mixture of calcium oxide or calcium hydroxide with sodium hydroxide.

SODIUM SULPHIDE, ANHYDROUS. A yellow or reddish-coloured solid having a strong odour. It is hygroscopic and oxidizes spontaneously on contact with air. Spontaneous ignition may occur in material improperly packed.

SOLID. Dangerous goods, other than gases, that do not meet the definition of liquid.

SOLUTION. Solution means any homogeneous liquid mixture of two or more chemical compounds or elements that will not undergo any separation under conditions normal to transport.

SOLVENTS. Substances capable of dissolving other substances to form a uniformly dispersed mixture or solution. Examples of organic solvent groups are esters, ethers, ketones, amines and nitrated and chlorinated hydrocarbons. Many solvents are flammable and toxic to varying degrees.

SOUNDING DEVICES, EXPLOSIVE. Articles consisting of a charge of detonating explosive. They are dropped from ships and function when they reach a predetermined depth or the sea bed.

✱ **SPECIAL ARRANGEMENTS.** (Radioactive Material Only). Those provisions approved by the competent authority under which a consignment, which does not satisfy all the applicable requirements of these Regulations may be transported. For international shipments of this type multilateral approval is required.

✱ **SPECIAL-FORM RADIOACTIVE MATERIAL.** (Radioactive Material Only). Either an indispersible solid radioactive material or a sealed capsule containing radioactive material that must be so manufactured that it can be opened only by destroying the capsule.

✱ **SPECIFIC ACTIVITY.** (Radioactive Material Only). The activity of the radionuclide per unit weight of that nuclide. The specific activity of a material in which radionuclides

are essentially uniformly distributed is the activity per unit weight of that material.

SPONTANEOUS IGNITION TEMPERATURE. The lowest temperature at which a substance will ignite spontaneously without an external source of ignition.

STABILIZED. Means that the substance is in a condition that precludes uncontrolled reaction. This may be achieved by methods such as the addition of an inhibiting chemical, degassing the substance to remove dissolved oxygen and inerting the air space in the package, or maintaining the substance under temperature control.

STATE OF DESTINATION. The country (State) in the territory of which the consignment is finally to be unloaded from an aircraft.

STATE OF ORIGIN. The country (State) in the territory of which the consignment is to first be loaded on an aircraft.

STATE OF REGISTRY. The country (State) on whose register the aircraft is entered.

STATE OF THE OPERATOR. The country (State) in which the operator's principal place of business is located or, if there is no such place of business, the operator's permanent residence.

STORES (SUPPLIES).

(a) **STORES (SUPPLIES) FOR CONSUMPTION.** Goods, whether or not sold, intended for consumption by the passengers and the crew on board aircraft and goods necessary for the operation and maintenance of aircraft; and

(b) **STORES (SUPPLIES) TO BE TAKEN AWAY.** Goods for sale to the passengers and the crew of aircraft with a view to being landed.

Items that meet classification as dangerous goods and which are transported in accordance with 2.5.2 are considered as "cargo".

Note:

The word "landed" in this context generally means duty free or tax free goods that will be declared to Customs by passengers and crew and which may be subject to excise duty.

SUBSTANCES, EXPLOSIVE, VERY INSENSITIVE (SUBSTANCES, EVI) N.O.S.. Substances that present a mass explosion hazard but which are so insensitive that there is very little probability of accidental initiation or of transition from burning to detonation under normal conditions of transport and which have passed Test Series 5.

SULPHURIC ACID, FUMING. Is sulphuric acid in which an excess of sulphur trioxide has been dissolved. It is more corrosive than ordinary sulphuric acid and evolves toxic fumes whilst ordinary sulphuric acid does not.

SULPHURIC ACID, SPENT. Is sulphuric acid usually of high concentration, which has been used for chemical processes and contains residual organic matter.

✱ **SURFACE CONTAMINATED OBJECT (SCO).** (Radioactive Material Only). A solid object which is not itself radioactive but which has radioactive material distributed on its surface.

T

TANK. A tank container, portable tank, a road tank vehicle, a rail tank wagon or a receptacle intended to contain solids, liquids, or gases and has a capacity of not less than 450 litres when used for the transport of gases as defined in 3.2.1.1.

Note:

These Regulations do not permit the use of a tank for the transport of radioactive material by air.

TEST CONDITION. Means the physical conditions imposed on the sample during the process of testing.

TEST PRESSURE. The required pressure applied during a pressure test for qualification or requalification.

TORPEDOES. Articles containing an explosive or non-explosive propulsion system and designed to be propelled through water. They contain an inert head or a warhead. The term includes:

- Torpedoes, liquid-fuelled, with inert head;
- Torpedoes, liquid-fuelled, with or without bursting charge;
- Torpedoes, with bursting charge.

TOTAL CONTENTS. (Explosive Material Only). Means such a substantial proportion that the practical hazard should be assessed by assuming simultaneous explosion of the whole of the explosive content of the load or package.

TRACERS FOR AMMUNITION. Sealed articles containing pyrotechnic substances, designed to reveal the trajectory of a projectile.

TRAINING COURSE. A series of lessons to teach the knowledge and competencies for a specific job function. The typical taught curriculum for dangerous goods personnel, mainly conducted through classroom training or e-learning.

TRAINING PROGRAM. The end-to-end process that consists of all competency-based training components. These include the training needs analysis, course materials, assessment materials, training records and instructor qualifications.

- ⚠ **TRANSPORT INDEX (TI).** (Radioactive Material Only). A single number assigned to a package, overpack or freight container or to unpackaged LSA-I, SCO-I or SCO-III to provide control over radiation exposure.

Note:

Unpackaged LSA-I, SCO-I or SCO-III material are not permitted in air transport.

TRAYS. (Explosive Material Only). Sheets of metal, plastics, wood, fibreboard or other suitable material which are placed in the Inner, Intermediate or Outer packaging and achieve a close-fit in such packaging. The surface of the tray may be shaped so that packagings or articles can be inserted, held secure and separated from each other.

- △ **TUBE.** (Not permitted for air transport.) A pressure receptacle of seamless or composite construction having a water capacity exceeding 150 L but not more than 3,000 L.

TURBINE ENGINES. Generic term used for turbine engines fuelled by flammable liquid, flammable gas or other combustible fuels. They may power fixed wing aircraft, rotorcraft, hovercraft (cushion craft), marine vessels, land vehicles, pumps and power generating plants.

TURPENTINE SUBSTITUTES. A petroleum distillate which might contain some aromatic components and which usually has a flash point of approximately 40°C. White spirit is a synonym for turpentine substitute.

U

ULLAGE. Amount by which a container, or a receptacle, falls short of being full.

Note:

Ullage is required to allow for the expansion of the contents.

UNCOMPRESSED GAS. Gas at a pressure not exceeding ambient atmospheric pressure at the time the containment system is closed.

UNECE. The United Nations Economic Commission for Europe (UNECE, Palais des Nations, 8, 14 avenue de la Paix, CH-1211 Geneva 10, Switzerland).

- ⚠ **UNILATERAL APPROVAL.** (Radioactive Material Only). An approval of a design which is required to be given by the competent authority of the State of origin of the design only.

- ⚠ **UNIRRADIATED THORIUM.** Thorium containing not more than 10^{-7} gram of uranium-233 per gram of thorium-232.

- ⚠ **UNIRRADIATED URANIUM.** Uranium containing not more than 2×10^3 Bq of plutonium per gram of uranium-235, not more than 9×10^6 Bq of fission products per gram of uranium-235 and not more than 5×10^{-3} g of uranium-236 per gram of uranium-235.

UNIT LOAD DEVICE. Any type of freight container, aircraft container, aircraft pallet with a net, or aircraft pallet with a net over an igloo.

Notes:

1. *An overpack is not included in this definition.*
2. *A freight container for radioactive material is not included in this definition.*

⊗

UN NUMBER. The four-digit number assigned by the United Nations Committee of Experts on the Transport of Dangerous Goods and on the Globally Harmonized System of Classification and Labelling of Chemicals to identify an article or substance or a particular group of articles or substances. (The prefix "UN" must always be used in conjunction with these numbers).

A

U
to
Z

☛ URANIUM—NATURAL, DEPLETED, ENRICHED.

- Natural uranium. Uranium (which may be chemically separated) containing the naturally occurring distribution of uranium isotopes (approximately 99.28% uranium-238 and 0.72% uranium-235 by mass).
- Depleted uranium. Uranium containing a lesser mass percentage of uranium-235 than in natural uranium.
- Enriched uranium. Uranium containing a greater mass percentage of uranium-235 than 0.72%.

In all cases, a very small mass percentage of uranium-234 is present.

V

VRI CODE. The distinguishing sign, consisting of one to three letters in capital Latin letters, of vehicles in International traffic as notified to the United Nations in accordance with the Geneva Convention on Road Traffic of 1949 or the Vienna Convention on Road Traffic of 1968 (See Appendix D.1 and D.2).

W

WARHEADS. Articles consisting of detonating explosives. They are designed to be fitted to a rocket, guided missile or torpedo. They may contain a burster or expelling charge or bursting charge. The term includes:

- Warheads, rocket, with burster or expelling charge;
- Warheads, rocket, with bursting charge;
- Warheads, torpedo, with bursting charge.

WATT-HOUR RATING. Expressed in Watt-hours, the Watt-hour rating of lithium ion battery is calculated by multiplying the rated capacity in ampere-hours by its nominal voltage.

WEIGHT. The force at which a body is attracted towards the earth and is equal to the mass multiplied by the acceleration due to gravity. For practical purposes, mass and weight are used interchangeably in these Regulations.

WHITE ASBESTOS. —see ASBESTOS.

△ WORKING PRESSURE. Either

- for a compressed gas, the settled pressure at a reference temperature of 15°C in a full pressure receptacle;
- for UN 1001, **Acetylene, dissolved**, the calculated settled pressure at a uniform reference temperature of 15°C in an acetylene cylinder containing the specified solvent content and the maximum acetylene content; or
- for UN 3374 **Acetylene, solvent free**, the working pressure which was calculated for the equivalent cylinder for UN 1001 **Acetylene, dissolved**.

Z

ZIRCONIUM SUSPENDED IN A FLAMMABLE LIQUID. Consists of very finely divided metallic zirconium which is usually suspended in some highly volatile and flammable liquid. If spilled, the material is liable to self-ignition and therefore can be shipped only in very limited quantities when specially packed.